Abdullah Al Fahad

AI, Multilingual NLP, and Computer Vision Researcher

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RESEARCH INTEREST & MOTIVATION

Natural Language Processing (NLP), Multilingual NLP, Large Language Model (LLM) Formal & Syllogistic Reasoning, Parameter-Efficient Fine-Tuning (QLoRA/LoRA), Explainable AI (LIME), Computer Vision, Vision Learning, and Dense Embedding Representations are among my core research interests.

I am a final-year (4th year) undergraduate student in Computer Science and Engineering at **Southeast University** with a strong academic record (**CGPA: 3.88/4.00** across 132 completed credits, achieving a perfect **4.00/4.00 SGPA** in my most recent semester). I have active research experience in Natural Language Processing, Large Language Models, and Explainable AI, and have authored **two accepted research papers as First Author** investigating **CrossCoT-QLoRA-Qwen3** layer optimization, **LIME** post-hoc explainability, and the disentanglement of semantic content plausibility from formal deductive logic in LLMs. I am also scheduled to attend and present my first-author work at the upcoming conference. Building on this foundation, I am actively expanding my research into **AI, vision learning, and multimodal embedding representations**, and I am highly motivated to pursue an **AI/NLP Research Internship** and a **fully funded MSc or PhD degree abroad** to develop trustworthy, interpretable, and generalizable AI systems.

EDUCATION

Summer 2023 – Present
(4th Year / Final Year)

Bachelor of Science (B.Sc.) in Computer Science and Engineering (CSE) – Department of Computer Science and Engineering, Southeast University, Dhaka, Bangladesh.

- **CGPA: 3.88 / 4.00** (Completed **132 Credits** across 10 semesters); **SGPA 4.00 / 4.00** in 10th Semester (Summer 2026) and 3rd Semester (Spring 2024); **SGPA 3.95 / 4.00** in 2nd Semester (Fall 2023).
- Consistently maintained merit-level academic standing (≥ 3.85 SGPA in 9 out of 10 completed semesters).
- **Core Coursework:** Artificial Intelligence, Machine Learning, Natural Language Processing, Computer Vision, Data Structures & Algorithms, Object-Oriented Programming (C++, Java), Database Management Systems (SQL), Web Design & Software Development.

RESEARCH & TECHNICAL EXPERIENCE

2025 – Present

Undergraduate AI & NLP Researcher, Department of Computer Science and Engineering, Southeast University – Dhaka, Bangladesh.

- Lead first-author research on **Multilingual NLP**, **LLM syllogistic and formal reasoning**, 4-bit quantized parameter-efficient fine-tuning (**QLoRA** on **Qwen3**), and post-hoc explainability (**LIME**).
- Designed Cross-lingual Chain-of-Thought (**CrossCoT**) prompting and layer-level optimization strategies to disentangle real-world semantic plausibility bias from deductive logical validity.
- Conducting ongoing research into **Computer Vision, Vision Learning, and Dense Embedding Representations**, focusing on visual feature learning, embedding space geometry, and cross-modal alignment.

2023 – Present

Software, Computer Vision & Data Science Developer, Department of Computer Science and Engineering, Southeast University – Dhaka, Bangladesh.

- Developed computer vision pipelines (human detection), AI-assisted web applications, and quantitative data analysis workflows using **Python**, **Pandas**, **Matplotlib**, and **PyCharm**.
- Engineered object-oriented software systems, embedded C++ control systems, relational databases (**SQL**), and responsive web interfaces using **C**, **C++**, **Java**, and modern **Web Design** tools.

PUBLICATIONS (GitHub; LinkedIn)

Accepted Peer-Reviewed Conference & Research Papers (First Author: 2 out of 2):

1. **Fahad, A. A.** *(First Author)*, et al. **(2026)**. Layer Optimization Leveraging CrossCoT-QLoRA-Qwen3 Framework with LIME Explainability for Multilingual Disentangling Content and Formal Reasoning in LLMs. *Accepted in Peer-Reviewed Conference Proceedings (2026)*. **[Accepted | Scheduled Conference Presenter]**
   - Proposed a layer-optimized **QLoRA** fine-tuning framework on the **Qwen3** architecture integrated with Cross-lingual Chain-of-Thought (**CrossCoT**) reasoning to disentangle semantic content bias from formal deductive validity across multilingual benchmarks.
   - Integrated **LIME** explainability to perform token-level attribution analysis and audit model reasoning faithfulness across valid/invalid and plausible/implausible syllogistic conditions.

2. **Fahad, A. A.** *(First Author)*, et al. **(2026)**. Plausibility Wins Over Validity: QLoRA Explainable CrossCoT in LLMs Prioritizes Content Over Logic in Syllogistic Reasoning. *Accepted in Peer-Reviewed Conference Proceedings (2026)*. **[Accepted]**
   - Investigated the "content effect" (belief bias) in quantized LLMs, providing empirical and explainability-driven evidence of how pre-trained semantic priors override formal deductive logic during syllogistic reasoning, and evaluated CrossCoT/QLoRA mitigation techniques.

SELECTED RESEARCH & TECHNICAL PROJECTS

1. **CrossCoT-QLoRA-Qwen3 & LIME Explainability Framework (Python, Qwen3, QLoRA, LIME, Pandas, Matplotlib):** Developed modular training, layer-sensitivity profiling, and LIME attribution pipelines for evaluating formal vs. content reasoning in multilingual LLMs.
2. **Computer Vision Human Detection Pipeline (GitHub: Human_Detection):** Implemented visual feature extraction and automated human detection workflows in Python and Jupyter Notebook utilizing Computer Vision techniques.
3. **StudyMate AI & Applied AI Implementations (GitHub: Studymateai | GitHub: Ai):** Engineered AI-powered study assistance web software and core Artificial Intelligence algorithms using Python, TypeScript, and modern Web Design principles.
4. **Wi-Fi Controlled Embedded System & Enterprise Network Architecture (GitHub: Rc-car | GitHub: HMS):** Built C++ wireless embedded control software and designed a multi-VLAN hospital network architecture (OSPF, VLSM, DHCP/DNS, NAT/PAT).

HONORS, ACHIEVEMENTS & ACADEMIC TRAJECTORY

1. **First-Author Undergraduate Research Distinction (2026):** Authored **2 accepted research papers** as First Author on LLM reasoning, Multilingual NLP, QLoRA, and Explainable AI during undergraduate studies.
2. **Selected Conference Presenter (2026):** Scheduled to attend and present the accepted first-author paper on the *CrossCoT-QLoRA-Qwen3 Framework*.
3. **Consistent High Academic Distinction (CGPA 3.88 / 4.00 across 132 Credits):** Achieved a perfect **4.00 / 4.00 SGPA** in both **Summer 2026** and **Spring 2024**, and **3.95 / 4.00 SGPA** in **Fall 2023**.

10-Semester Undergraduate GPA Progression at Southeast University (132 Credits Completed):

Term	Sum'23	Fall'23	Spr'24	Sum'24	Fall'24	Spr'25	Sum'25	Fall'25	Spr'26	Sum'26	Overall
Credits	12	14	14	14	13	14	13	14	12	12	**132**
SGPA	3.81	3.95	**4.00**	3.66	3.87	3.88	3.85	3.88	3.90	**4.00**	–
CGPA	3.81	3.88	3.93	3.86	3.86	3.86	3.86	3.86	3.86	**3.88**	**3.88**

TECHNICAL SKILLS

1. **Artificial Intelligence & Research:** Artificial Intelligence (AI); Natural Language Processing (NLP); Multilingual NLP; Computer Vision; Vision Learning & Dense Embeddings; Large Language Models (LLMs); Parameter-Efficient Fine-Tuning (QLoRA / LoRA); CrossCoT Reasoning; Explainable AI (LIME); Research Skills.
2. **Programming Languages:** Python (Programming Language), C (Programming Language), C++, Java, SQL.
3. **Data Analysis & Visualization:** Data Analysis, Pandas (Software), Matplotlib, Jupyter Notebook, Statistical & Quantitative Evaluation.
4. **Software Development & Tools:** Software Development, Web Design, PyCharm, Git & GitHub, LaTeX.

CONFERENCES, WORKSHOPS & SEMINARS ATTENDED

2026 — **Conference Presenter & Delegate** – Presenting the accepted first-author paper *"Layer Optimization Leveraging CrossCoT-QLoRA-Qwen3 Framework with LIME Explainability for Multilingual Disentangling Content and Formal Reasoning in LLMs"*.

2023 – 2026 — **AI/ML Research Seminars, Competitive Programming & Software Development Workshops** – Conducted by the Department of Computer Science and Engineering and Southeast Computer Club (SECC), Southeast University, Dhaka, Bangladesh.

PROFESSIONAL & EXTRACURRICULAR AFFILIATION

Member, **Southeast Computer Club (SECC)**, Department of Computer Science and Engineering, Southeast University, Dhaka, Bangladesh *(2023 – Present)*.